

rsync Cheat Sheet

Copy and sync files between folders or servers, sending only what changed.

Using it

```
rsync [options] source/ destination/ ← copy from source to destination
rsync -av --dry-run src/ backup/ ← show what would be copied, change nothing
rsync -avz src/ mark@web1:/srv/app/ ← copy to another server over SSH
```

Option	What it does	Option	What it does
-a	Archive: keeps permissions, times, links	-n	Dry run: only show what would happen
-v	List each file as it's copied	--delete	Delete destination files gone from source
-z	Compress while sending (slow links)	--exclude='*.log'	Skip matching files (repeatable)
-P	Progress bar + keep partial files to resume	-i	Show a change code per file (see below)
-h	Human-readable numbers (KB, MB)	-e 'ssh -p 2222'	Use SSH on a different port

The trailing slash rule

A slash at the end of the **source** means "the contents of"; no slash means "the folder itself".

```
rsync -a photos/ backup/ ← copies what's INSIDE photos (a slash on backup/ changes nothing)
    backup/a.jpg
    backup/b.jpg

rsync -a photos backup/ ← copies the photos FOLDER itself
    backup/photos/a.jpg
    backup/photos/b.jpg
```

Over SSH

Command	Does
rsync -av site/ mark@web1:/var/www/site/	Pushes a local folder up to a server
rsync -av mark@web1:/var/log/app/ ./logs/	Pulls a folder down from a server
rsync -av -e 'ssh -p 2222' src/ mark@web1:dst/	Uses a non-standard SSH port (path is relative to home)

Reading the output

```
$ rsync -avhi --delete src/ backup/
cd+++++++ docs/ ← new folder created
>f+++++++ docs/new.txt ← new file copied (+ = everything is new)
>f.st..... report.pdf ← updated: s = size changed, t = time changed
.f....g.... notes.txt ← only the group changed; no data sent
*deleting old.txt ← removed from backup (because of --delete)
sent 2.10M bytes received 1.2K bytes ...
total size is 512.4M speedup is 243.70 ← high speedup = most files were already there
```

First letter: > = copied here, < = copied to remote, **c** = created, . = no data sent. Second: **f** = file, **d** = folder, **L** = link.

Handy recipes

Command	Does
rsync -a /home/ /mnt/backup/home/	Backs up /home (only new and changed files)
rsync -a --delete src/ mirror/	Makes mirror an exact copy (deletes extras!)
rsync -avP big.iso mark@host:/tmp/	Copies a big file; run again to resume if it breaks
rsync -av --exclude='.git' ./ host:app/	Deploys code, skipping the .git folder
rsync -avn --delete src/ dst/	Previews what a mirror would change

Remember: always do a dry run (**-n**) before using **--delete**. A swapped source/destination or a missing slash can wipe the wrong side.